

Nonlinear Systems : Chapter 4

Describing Function Analysis

- Aim : **To predict the limit cycles**
- Method : Generalization of Frequency domain Methods
- This method has the following advantages :

- **i** : Results in algebraic, but complex, analysis
- **ii** : Uses **phasor** type analysis
- **iii** : Suitable for graphical interpretation
- **iv** : Suitable for experimental analysis

- **Example** : Van der Pol Oscillator

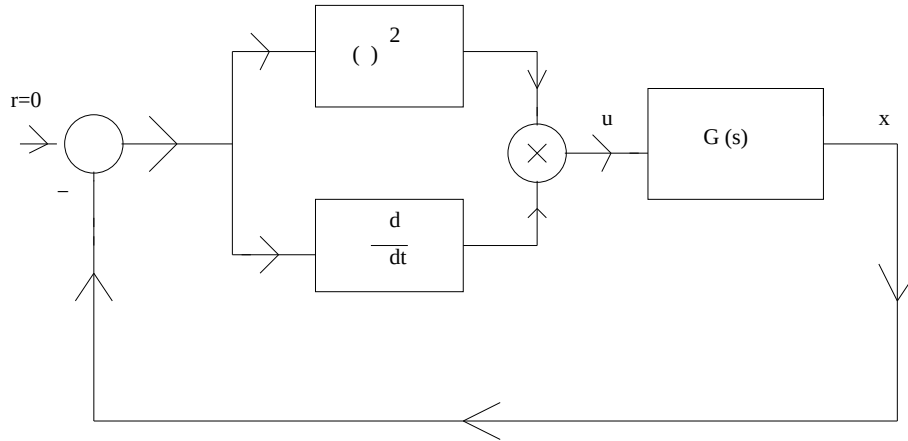
- $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$

- It is well known that this system has a unique limit cycle for large μ

- $\ddot{x} - \mu\dot{x} + x = -\mu x^2\dot{x}$

- Set $u = -x^2\dot{x} \implies \ddot{x} - \mu\dot{x} + x = \mu u$

- Take Laplace transform $\implies G(s) = \frac{X(s)}{U(s)} = \frac{\mu}{s^2 - \mu s + 1}$



- Assume that the system is in the limit cycle. **Question** : What is the amplitude and the fundamental frequency of the oscillation?

- Consider the signal in front of the Nonlinear block. Call it v . In the present case, $v = -x$

- Set $v = A \sin \omega t \implies u = v^2 \dot{v} = A^2 (\sin \omega t)^2 A \omega \cos \omega t$

- $\sin^2 \theta = 0.5(1 - \cos 2\theta)$, $\cos a \cos b = 0.5[\cos(a + b) + \cos(a - b)]$

- $\implies u = 0.25 A^3 \omega (\cos \omega t - \cos 3\omega t)$

- Assume that the linear block acts like a **low pass filter** $\implies$ neglect 3ω frequency

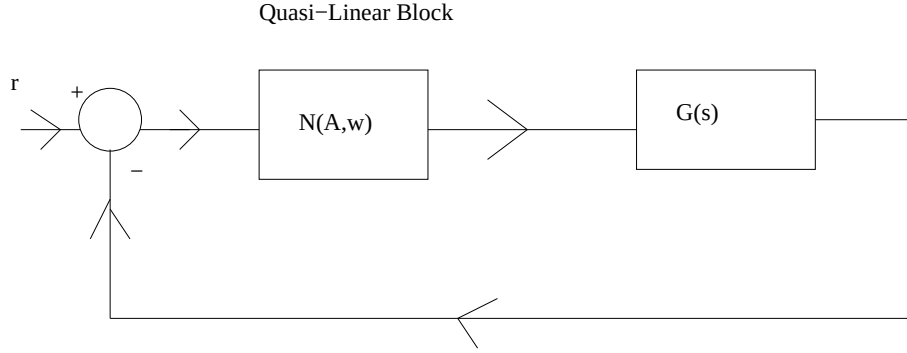
- $v = A \sin \omega t \implies u = 0.25 A^3 \omega \cos \omega t$

- Remember Phasor Analysis : $A \cos(\omega t + \phi) = \Re\{A e^{j\phi} e^{j\omega t}\}$

- $y(t) = A \cos(\omega t + \phi) \iff$ Phasor Y of $y(t)$ is $Y = A e^{j\phi}$

- $v = A \sin \omega t = A \cos(\omega t - 0.5\pi) \iff$ Phasor V is $V = A e^{-j0.5\pi} = -jA$

- $u = 0.25 A^3 \omega \cos \omega t \iff$ Phasor U is $U = 0.25 A^3 \omega$



- For **phasor analysis**, the NonLinear (NL) Block can be replaced by **Quasi-Linear Block**

- The transfer function of this block, $N(A, \omega)$ is called the **describing function** of the NL block :

- $N(A, \omega) = \frac{\text{Output Phasor}}{\text{Input Phasor}} = \frac{U}{V} = \frac{0.25A^3\omega}{-jA} = 0.25A^2(j\omega)$

- $U = N(A, \omega)V$, $X = G(j\omega)U$, $V = -X$

- **Basic Equation :**

- $(1 + N(A, \omega)G(j\omega))X = 0 \iff 1 + N(A, \omega)G(j\omega) = 0$

- Note that this equation is **complex**; Hence we can find the two unknowns : A (magnitude of the possible oscillation), and ω (frequency of the possible oscillation)

- For the Van der Pol oscillator case :

- $1 + 0.25A^2(j\omega) \frac{\mu}{(j\omega)^2 - \mu(j\omega) + 1} = 0$

- $(1 - \omega^2) + j\mu\omega(0.25A^2 - 1) = 0 \implies \omega = 1$, $A = 2$

- Hence a possible oscillation : $x(t) = 2 \sin t$

- **Stability Analysis**

- Set $j\omega \rightarrow s$:

- $1 + 0.25A^2 s \frac{\mu}{s^2 - \mu s + 1} = 0 \implies s^2 + \mu(0.25A^2 - 1)s + 1 = 0$

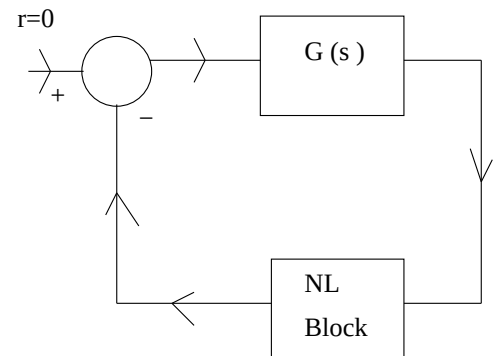
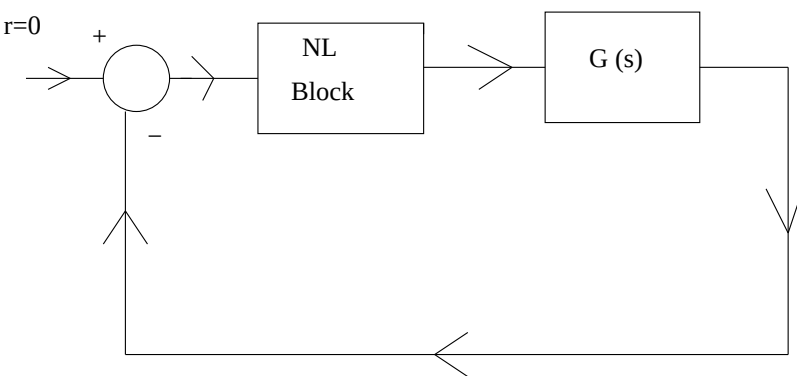
- $s_{1,2} = -0.5\mu(0.25A^2 - 1) \pm \sqrt{0.25\mu^2(0.25A^2 - 1)^2 - 1}$

- If $A \uparrow 2 \implies \Re\{s_{1,2}\} < 0 \implies$ System becomes stable $\implies A \downarrow$

- If $A \downarrow 2 \implies \Re\{s_{1,2}\} > 0 \implies$ System becomes unstable $\implies A \uparrow$

- **Conclusion :** This oscillation is **stable** (i.e. can be sustained)

- Exactly the same analysis is applied to the following type of systems :



- **Typical Procedure**

- **i** : Assume $v = A \sin \omega t$ as input to the NL Block

- **ii** : Find the output u of the NL Block. Consider only the **first harmonic**

- **iii** : Find the describing function $N(A, \omega)$ as the ratio of the output phasor U and input phasor V

- **iv** : Solve the main equation $1 + N(A, \omega)G(j\omega) = 0$, find A and ω

- **Typical example :**

- $x^{(n)} + a_{n-1}x^{(n-1)} + \dots + a_0x + f(x) = 0$

- $\implies$ NL Block : $u = f(v)$, Linear Block : $G(s) = \frac{1}{s^n + a_{n-1}s^{n-1} + \dots + a_0}$

- This could also be extended to the case where f depends on the derivatives of x as well

- Example : Van der Pol oscillator :

- $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$

- This could also be extended to the case where $r = r_0 \sin \omega t$ To predict the frequency domain behaviour of the closed loop system.

- **Basic Assumptions** (Simple Version)

- **i** : Only one nonlinear block.

- For simplification. Can be eliminated.

- **ii** : Nonlinearity is memoryless and time-invariant.

- First can be relaxed. Second is needed for Nyquist stability analysis.

- **iii** : Only the fundamental frequency at the output is considered.

- This is a fundamental assumption and cannot be relaxed. This is physically justified if the linear block is a **low-pass filter**.

- **iv** : Nonlinearity is **odd** (i.e. $f(e) = -f(-e)$).

- Simplifies the Fourier analysis.

- Assume that the input to the NL block is sinusoidal, i.e. $v(t) = A \sin \omega t$, and NL Block gives $u(t) = f(v(t))$.

- If f is single valued function, than $u(t)$ is also periodic with the same fundamental frequency

- $u(t) = f(A \sin \omega t) = f(A \sin \omega t + 2n\pi) = f(A \sin \omega(t + 2n\pi/\omega))$

- $u(t) = u(t + nT) \quad , \quad T = 2n\pi/\omega$

- As a result, we can express $u(t)$ as a Fourier series

- $$u(t) = 0.5a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

- Remember the **orthogonality** of sin and cos functions, i.e. : For $n \neq m$

- $$\int_{-\pi/\omega}^{-\pi/\omega} \sin n\omega t \sin m\omega t \, dt = \int_{-\pi/\omega}^{-\pi/\omega} \cos n\omega t \cos m\omega t \, dt$$
- $$= \int_{-\pi/\omega}^{-\pi/\omega} \sin n\omega t \cos m\omega t \, dt = 0$$

- Using this property, we easily obtain :

- $$\int_{-\pi/\omega}^{-\pi/\omega} u(t) \sin n\omega t \, dt = b_n \int_{-\pi/\omega}^{-\pi/\omega} (\sin n\omega t)^2 \, dt$$
- $$= \int_{-\pi/\omega}^{-\pi/\omega} 0.5(1 - \cos 2n\omega t) \, dt = b_n \pi / \omega$$

- $$b_n = \frac{\omega}{\pi} \int_{-\pi/\omega}^{-\pi/\omega} u(t) \sin n\omega t \, dt$$

- Coordinate change : $\omega t = \theta$

- $$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(A \sin \theta) \sin n\theta \, d\theta$$

- Similarly we have :

- $$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(A \sin \theta) \cos n\theta \, d\theta$$

- $$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(A \sin \theta) \, d\theta$$

- If f is **odd** $\implies a_0 = 0!$
- Fundamental terms $\implies u(t) = a_1 \cos \omega t + b_1 \sin \omega t$
- Phasor of $b_1 \sin \omega t \implies -jb_1$
- Phasor of $u(t) \implies a_1 - jb_1$
- Describing Function :
- $$N(A, \omega) = \frac{\text{Output Phasor}}{\text{Input Phasor}} = \frac{U}{V} = \frac{a_1 - jb_1}{-jA} = \frac{b_1 + ja_1}{A}$$